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Disks and Jets

Quasars

Most of the objects seen by radio telescopes are huge clouds of gas, clouds far larger than any star. But in the early 1960s a few tiny objects were found. Astronomers named these objects *quasars* for “quasi-stellar radio sources.”

In 1962 the Caltech astronomer Maarten Schmidt, looking through the world’s largest optical telescope on Palomar Mountain, discovered light coming from a quasar called 3C273. It looked like a bright star with a faint jet shooting out of it (Figure 9.1). This was weird!

When Schmidt split 3C273’s light into its various colors (as is sometimes done by sending light through a prism), he saw the set of spectral lines in the bottom of Figure 9.1. At first sight, these were unlike any spectral lines he had ever seen. But in February 1963, after a few months’ struggle, he realized the lines were unfamiliar simply